GHRaisoni International Skill Tech University, Pune

AI CHATBOT FOR COLLEGE

FACULTY REVISED SYSTEM ARCHITECTURE & DIAGRAM DOSSIER

✓ 3 Swimlanes (Student | Chatbot | Database) ✓ Database Student Record Fetching ✓ 4-Tier Attendance Detention Policy

1. Activity Diagram — 3-Swimlane Session Flow with Database Integration

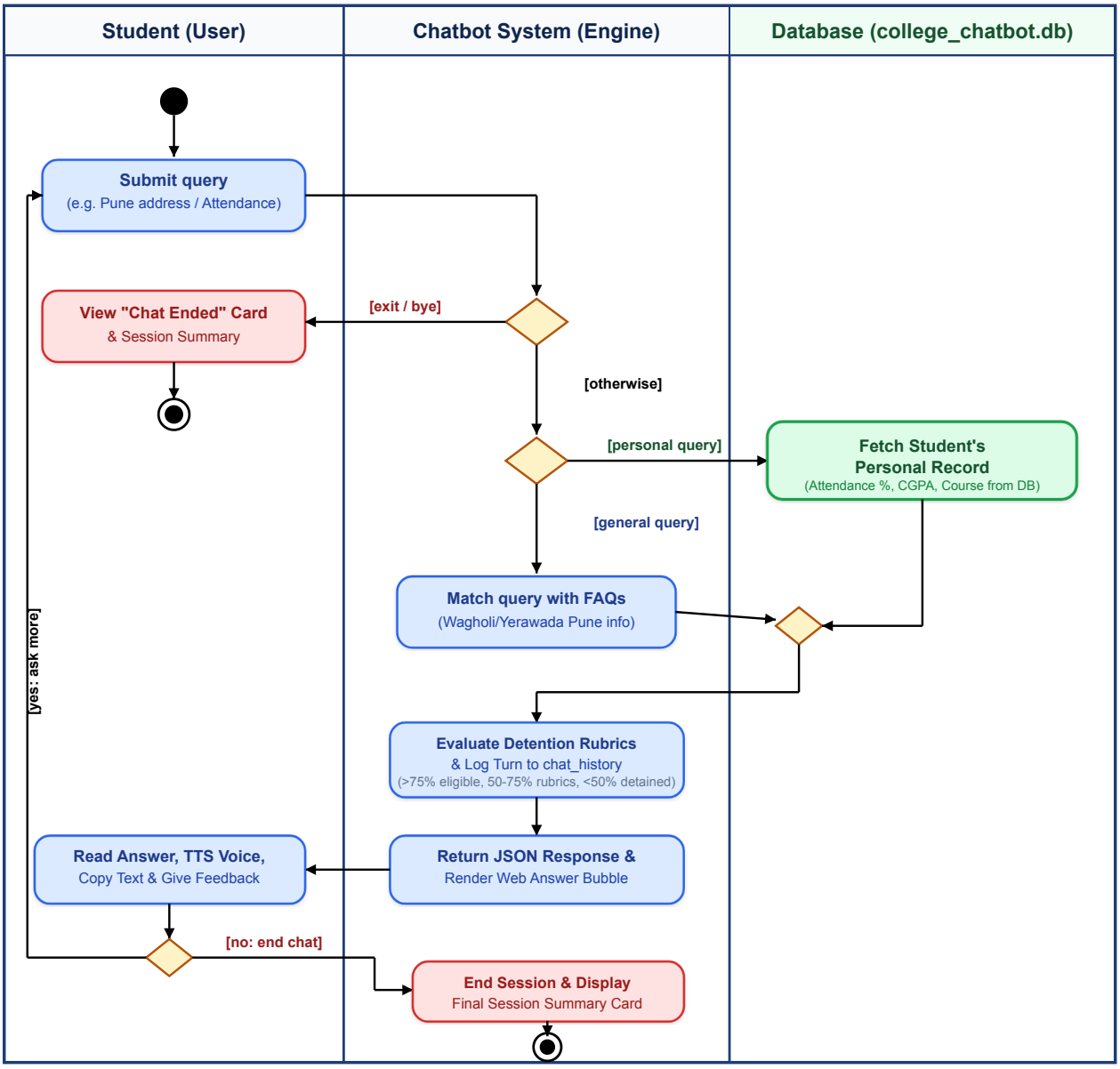


Fig 1: Activity Diagram — 3-Swimlane architecture (Student, Chatbot System, Database) showing explicit student personal record retrieval (attendance & CGPA), FAQ matching for Pune campuses (Wagholi & Yerawada), 4-tier detention policy evaluation, and controlled session loop with explicit End states.

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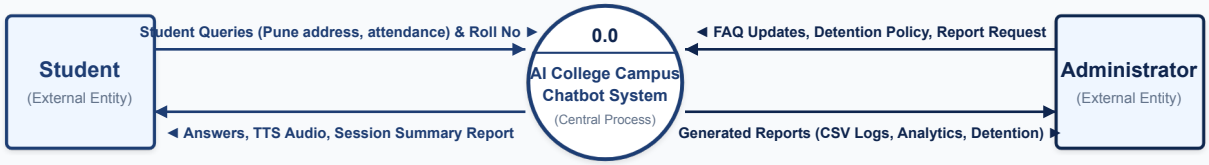
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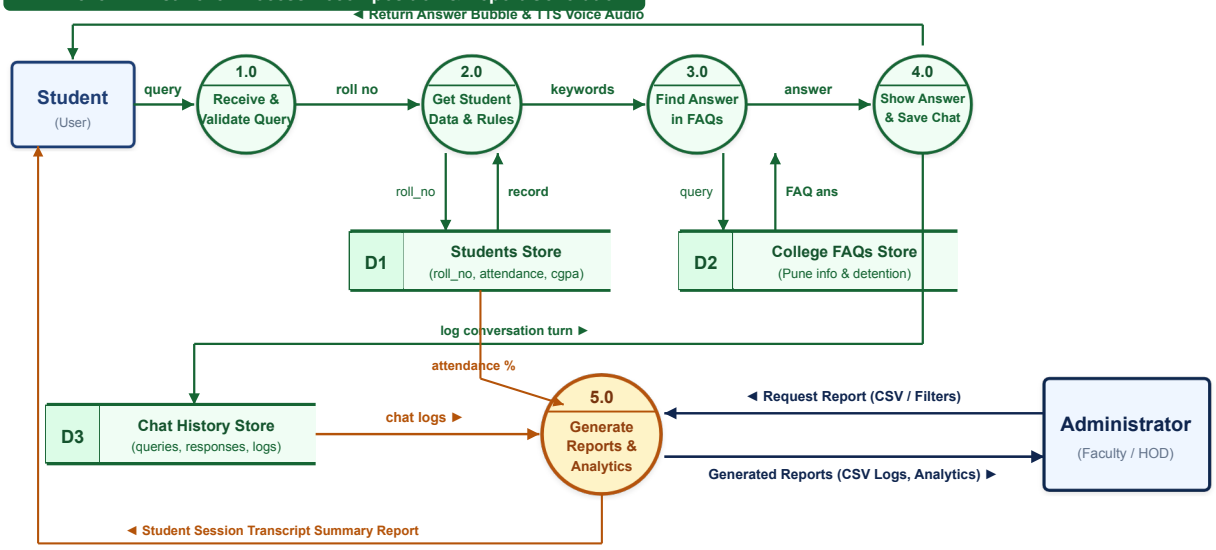
✓ Level 1 Process Decomposition (1.0 – 4.0) | ✓ 3 Data Stores (D1 Students, D2 FAQs, D3 Chat) | ✓ Official DFD Symbols Reference Table

2. Data Flow Diagram (DFD Level 0 Context & Level 1 Process Breakdown with Report Generation)

DFD Level 0: Context Diagram



DFD Level 1: First-Level Process Decomposition & Report Generation



Standard DFD Notation & Symbol Conventions (Gane-Sarson & DeMarco)

Symbol Shape	DFD Element	Standard Academic Definition	Implementation in Our Project
▭	External Entity	Origin/destination of data outside boundary (Square)	Student (User) & Administrator (Faculty/HOD)
◯	Process (Bubble)	Transforms input data into output data (Circle with ID) Calculates, retrieves, formats, or generates reports	1.0 to 4.0 Query Handlers & 5.0 Generate Reports Query answering & administrative CSV/analytics reporting
D ═	Data Store *(Open-Ended)*	Repository of data at rest (Open parallel lines) **MANDATORY RULE: Right side must be open; NEVER closed**	D1 Students, D2 College FAQs, D3 Chat History Persistent SQLite tables feeding search & report generation
→	Data Flow	Directed solid arrow showing data path & payload Transfers packets between entities, processes, and stores	query, record, FAQ answer, generated reports CSV Directional vectors with explicit noun-phrase labels

Fig 2: Data Flow Diagram (DFD) — Properly formatted Level 0 Context Diagram (System boundary with Student & Administrator) plus Level 1 Process Decomposition (1.0 to 4.0 Query Handlers & 5.0 Generate Reports & Analytics) with standard open-ended data stores (D1 Students, D2 FAQs, D3 Chat History) and formal academic DFD symbols reference table.

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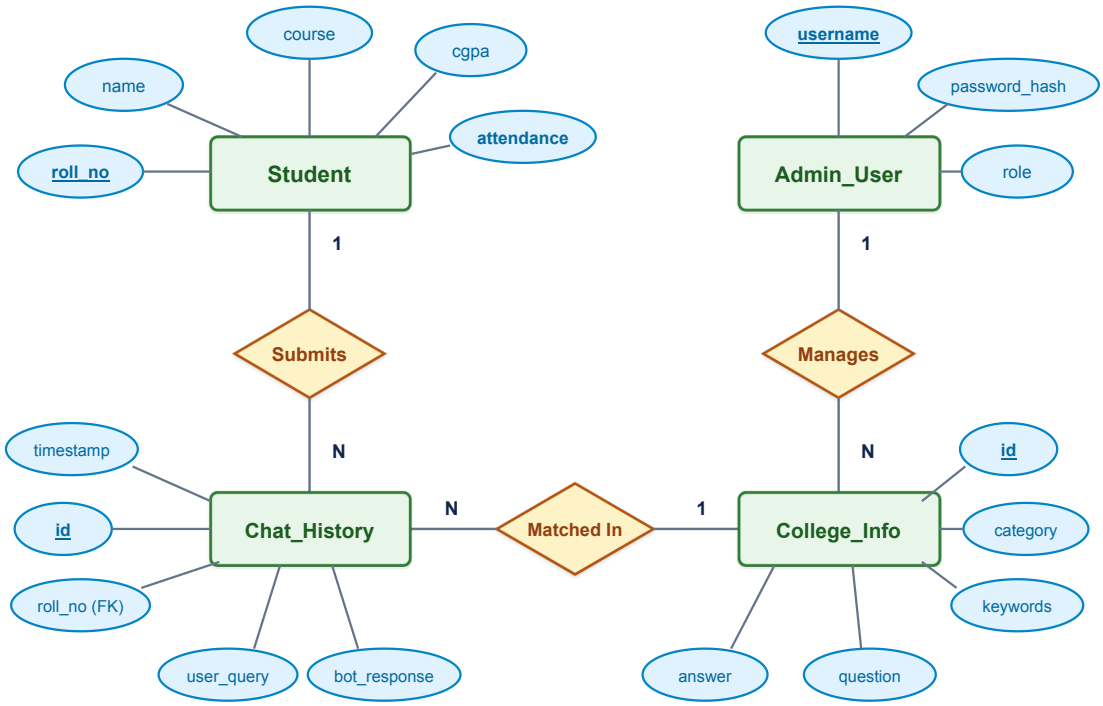
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✓ Classic Academic ER Notation | ✓ 4 Core Entities (Student, Admin, Chat, Info) | ✓ 3NF Normalization Defense & Legend Key

3. Entity-Relationship Diagram (ERD — Classic Notation) & Relational Schema

ER Diagram (Classic Notation): AI Chatbot for College



Schema & Project Knowledge Base Defense:

- **Student Entity:** Stores attendance %, CGPA & evaluates 4-tier detention policy.
- **College_Info Entity:** Contains Pune campuses (Wagholi & Yerawada) & exam rules.
- **Chat_History Entity:** Logs each turn with student roll_no FK, query & bot answer.
- **Admin_User Entity:** RBAC authentication for updating college FAQs & detention data.

Verified in 3NF Normalization: Zero transitive dependencies & lossless join preservation.

Key / Legend:

- Entity (Table)
- Attribute (Column) — id = Primary Key
- Relationship
- Link / Cardinality (1:N)

Fig 3: Entity-Relationship Diagram (ERD — Classic Notation) — 4 core entities (Student, Admin_User, Chat_History, College_Info) with attributes (PK underlined, FK indicated), cardinalities (1:N, N:1), and official symbol key matching faculty review specifications.